

Origin, fate and function of extraembryonic tissues during mammalian development

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Abstract

Extraembryonic tissues have pivotal roles in morphogenesis and patterning of the early mammalian embryo. Developmental programmes mediated through signalling pathways and gene regulatory networks determine the sequence in which fate determination and lineage commitment of extraembryonic tissues take place, and epigenetic processes allow the memory of cell identity and state to be sustained throughout and beyond embryo development, even extending across generations. In this Review, we discuss the molecular and cellular mechanisms necessary for the different extraembryonic tissues to develop and function, from their initial specification up until the end of gastrulation, when the body plan of the embryo and the anatomical organization of its supporting extraembryonic structures are established. We examine the interaction between extraembryonic and embryonic tissues during early patterning and morphogenesis, and outline how epigenetic memory supports extraembryonic tissue development.

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Introduction

During mammalian development, the fertilized egg gives rise not only to the embryo, which becomes the born individual, but also to distinct extraembryonic tissues that are intimately juxtaposed to it, together forming the ‘conceptus’. The evolution of extraembryonic tissues was a prerequisite to enable viviparous development in mammals. Their formation is dictated by morphogenetic processes that have been spatially and temporally finetuned through evolution, to balance reliability and efficacy of outcomes, working against constraints imposed by resource availability, gestational time and environmental variability. This balance is thought to be achieved through an intricate interplay between various molecular mechanisms – signalling pathways, transcriptional networks and epigenetic processes – to generate the conceptus in a timely and reliable manner.

Extraembryonic tissues form in concert with embryonic tissues, and reciprocal interactions between them ensure their coordinated development, to facilitate the transient roles of these tissues^{1,2}. Extraembryonic tissues are formed in rapid succession to each other during early development, ultimately giving rise to the fetal membranes (yolk sac and amnion) and the placenta, which are discarded at birth (Fig. 1a,b). They serve important functions beyond providing for the nutritional needs of the embryo, offering mechanical protection and immunological camouflaging during its development³. Extraembryonic tissues also orchestrate implantation into the uterus, help shape and pattern the embryo and act as a source of cells contributing to embryonic organs and as a site for early haematopoiesis⁴. Over the past decade, a growing number of single-cell molecular profiling datasets from various embryonic stages, and their experimental validation, has advanced our understanding of the genetic basis for the origin and fates of extraembryonic tissues^{5–9}. Epigenetic modifications are now known to form an increasingly integral part of the molecular description of extraembryonic tissues, revealing a larger role for this level of molecular regulation than previously appreciated⁶. Complementing transcriptional and translational regulation, epigenetic modifications

to both DNA and histone proteins contribute to the generation and maintenance of heterogeneities within the developing conceptus, with tissue-specific epigenetic information facilitating precision of gene regulation during embryonic development^{6,10}.

Revisiting the role of extraembryonic tissues is particularly timely, given continuing efforts at refining stem-cell-based in vitro models of mammalian embryos^{11–17}. Such models have improved in recent years through the incorporation of extraembryonic cell types^{14–16} and can further benefit from a better understanding of the role of extraembryonic tissues during development in utero. This will enable them in turn to serve as powerful models for investigating events that depend on extraembryonic tissues but are particularly difficult to address in vivo, such as implantation.

In this Review, we focus on extraembryonic structures before the formation of the placenta, a complex structure whose development and function have been reviewed elsewhere^{18–20}. We will discuss the origin and pivotal role of extraembryonic tissues in shaping and patterning the mammalian embryo, outlining the molecular mechanisms underpinning these processes, from the time they first emerge up until gastrulation. We will focus on studies from the mouse, a genetically tractable model that has been instrumental in advancing our understanding of the cellular and molecular control of these processes. We will outline the important epigenetic mechanisms that contribute to the development of these tissues and specifically discuss the interplay between signalling pathways and transcriptional regulation, dictating the cellular behaviours necessary for the development and functioning of extraembryonic tissues.

Ontogeny of extraembryonic tissues

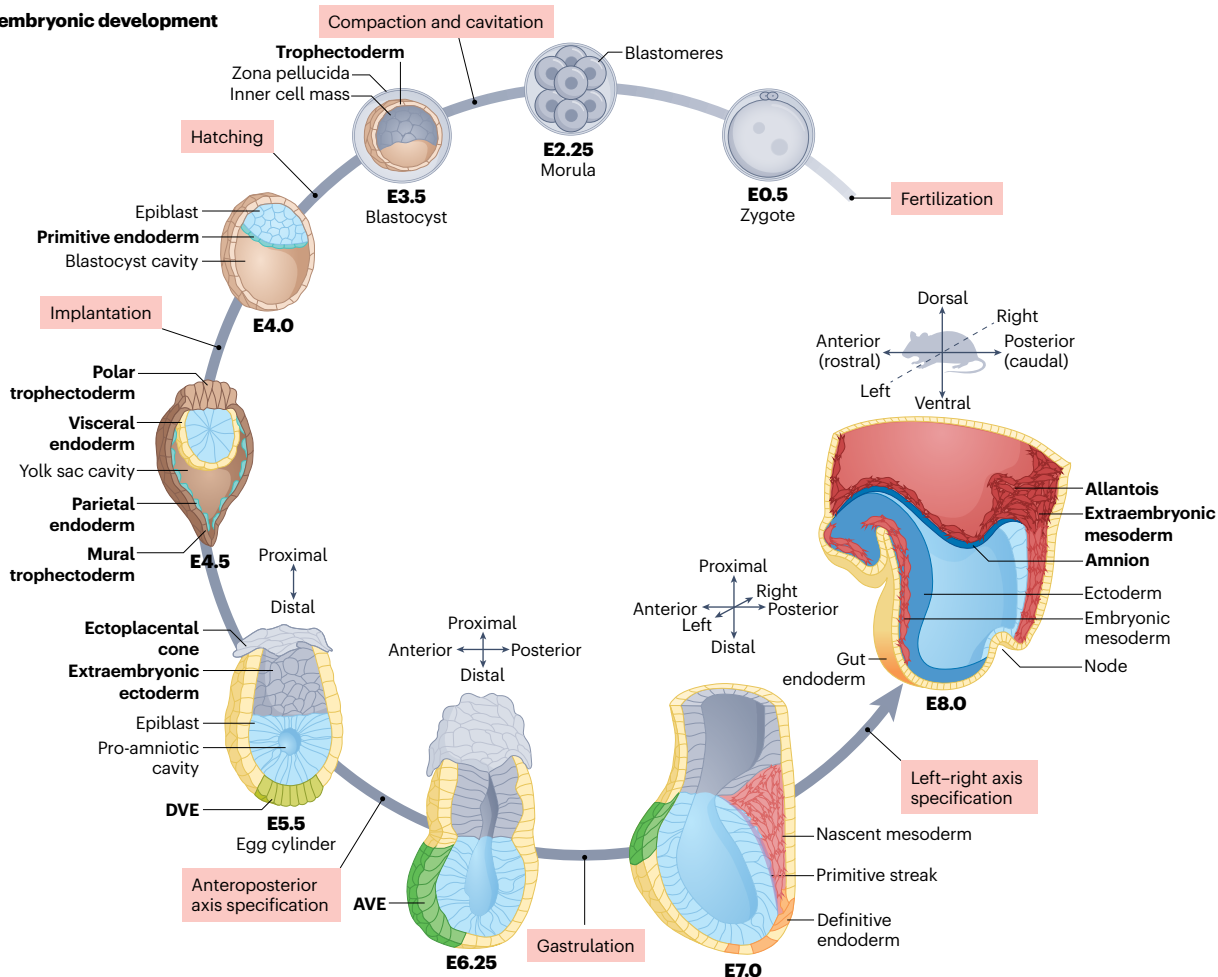
The distinctive viviparous and matrotrophic developmental strategies employed by eutherian (true placental) mammals demand a substantial prenatal maternal energy investment to ensure reproductive success. Extraembryonic tissues form the interface (placenta) as well as the barrier (fetal membranes) between the mother and the fetus^{21,22} (Fig. 1b).

Fig. 1 | The origin and organization of tissues during early mouse development.

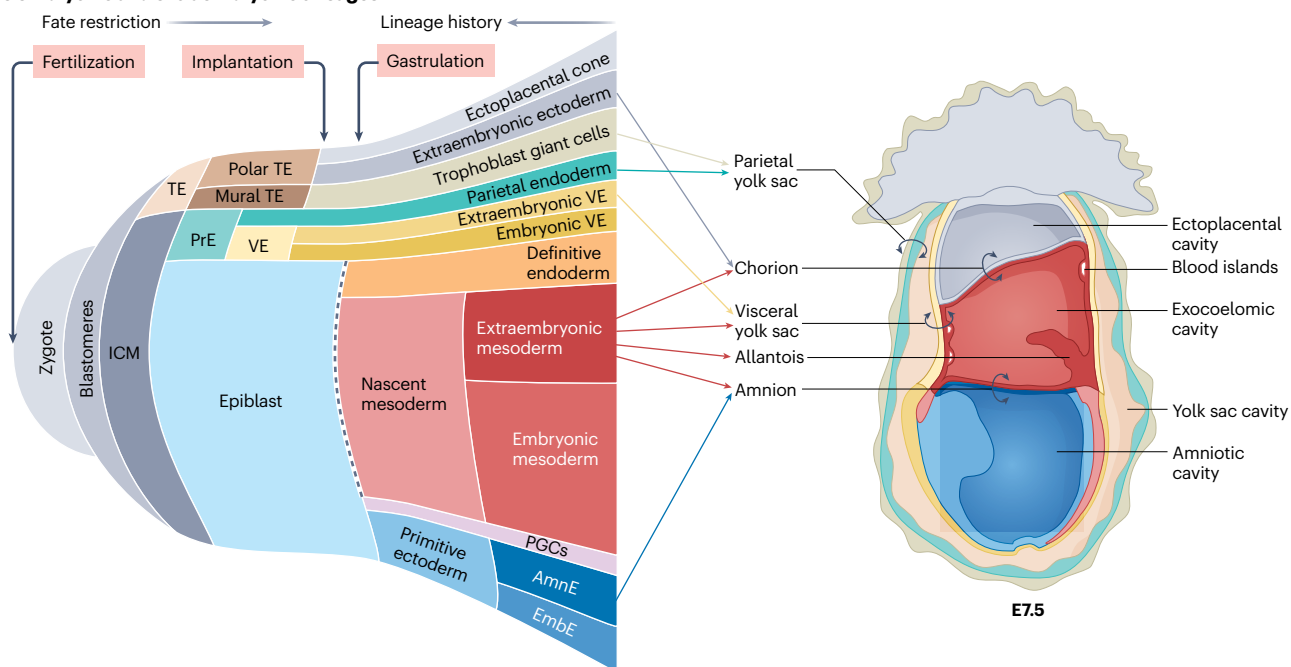
a, Schematic of development from fertilization to embryonic day (E) 8.5, showing where and when the extraembryonic tissues first emerge. The main morphogenetic events during early embryogenesis are shown. Following fertilization, cell division of the zygote leads to the formation of the morula, which consists of blastomeres. Subsequently, compaction and then cavitation result in the blastocyst, marked by the development of the first extraembryonic tissue – the trophectoderm (TE) – enclosing the inner cell mass (ICM). By the time the blastocyst hatches from the zona pellucida, the ICM is asymmetrically positioned within the prolate blastocyst, and subsequently differentiates into the epiblast and primitive endoderm (PrE). A subset of PrE cells undergo an epithelial–mesenchymal transition, generating the parietal endoderm. By implantation (E4.5), the embryo acquires an egg cylinder morphology with a distinct proximodistal axis relative to the point of implantation. The remaining PrE cells become the visceral endoderm (VE). Within the visceral endoderm, a subset of cells, the distal visceral endoderm (DVE), acquires a distinct columnar morphology. The DVE migrates from the distal tip to the prospective anterior region of the embryo, to give rise to the anterior visceral endoderm (AVE) and pattern the first definitive anatomical axis, the anteroposterior axis. The AVE restricts the formation of the primitive streak to the opposite end of the egg cylinder, where epiblast cells delaminate and give rise to mesoderm and endoderm. The extraembryonic mesodermal cells coordinate with the existing endodermal and ectodermal cells to generate various extraembryonic tissues. The extraembryonic tissue types that contribute to the fetal membranes and

placenta are shown in bold. For the stages beyond E5.5, the outermost cells that form the parietal yolk sac are not shown for the sake of clarity. Similarly, for E7 and beyond, the proximal most population of cells that make up the ectoplacental cone are not shown. The various anatomical axes and the order in which they emerge are indicated within the centre of the circle. **b**, Embryonic and extraembryonic tissues emerge sequentially from fertilization up until gastrulation during mouse development. Commitment of trophectoderm into polar and mural trophectoderm occurs in preparation for implantation. The former gives rise to the ectoplacental cone and extraembryonic ectoderm, whereas the latter forms the trophoblast giant cells. By contrast, commitment of epiblast derivatives to various embryonic and extraembryonic lineages happens following gastrulation through the primitive streak. The dashed vertical line represents the primitive streak, indicating the subsequent fate of cells that ingress through it. The cell types that come together to form the various extraembryonic membranes (parietal or primary yolk sac, visceral or secondary yolk sac, amnion) and placental components (chorion and allantois) and their anatomical locations in the embryo at E7.5 are shown on the right. The embryonic cavities formed separate the embryo from the extraembryonic tissues, compartmentalizing development and organizing the embryo along the three anatomical axes. The embryonic or extraembryonic status of any tissue can be determined by tracing its fate forward through developmental time (to the right) and assessing whether it contributes to embryonic structures or only extraembryonic tissues. AmnE, amniotic ectoderm; EmbE, embryonic ectoderm; PGC, primordial germ cell. Part **a** is adapted with permission from ref. 255, Elsevier; and part **b** is adapted with permission from ref. 2, Royal Society.

a Mouse embryonic development



b Mouse embryonic and extraembryonic lineages



Understanding mammalian extraembryonic tissues at a molecular and cellular level, in a developmental context and through an evolutionary lens (Box 1), enables us to better appreciate why these tissues are the settings for specific epigenetic mechanisms unique to mammals.

Developmental origin of extraembryonic tissues in the pre-implantation embryo

As implantation is a necessary step for viviparity in mammals, the first stages of development are dedicated to generating the cell types required for this process. The early embryo is a self-organizing system, in which the boundaries between tissues gradually become more discrete as the potential of their constituent cells gets progressively restricted²³. In the mouse, after three rounds of cleavage of the totipotent zygote, the resulting blastomeres show increased cell–cell contacts and undergo compaction, thereby forming the roughly spherical morula, composed of cells with apicobasal polarity^{24,25} by embryonic day (E) 2.25 (Fig. 1a). In subsequent divisions, cells start to internalize.

Box 1 | The evolutionary need for extraembryonic tissues in mammals

The evolution of extraembryonic tissues parallels the transition of life forms to explore new environmental niches and the novel reproductive strategies that arose to facilitate this^{21,256,257}. The yolk sac phylogenetically predates other extraembryonic membranes, serving to absorb nutrients from the yolk of the aquatic egg. The inverted yolk sac (with a maternal facing absorptive endoderm) of many eutherian embryos continues to serve such an accessory histotrophic function^{258,259}. However, with the vertebrate land invasion that led to the emergence of amniotes, it became necessary to reorganize the aquatic egg to cope with new modes of gas exchange and excretion, as well as put in place structures to prevent desiccation. Several new extraembryonic tissues evolved to meet these demands^{22,260}. The amnion offers mechanical protection to the embryo and the allantois helps contain fetal excretions, whereas the vascularized chorion, similar to the yolk sac, facilitates nutrient and gas exchange^{4,21,22}.

With the evolution of viviparity, existing extraembryonic tissues were repurposed to enable placentation. Although yolk sac placentation as seen in marsupials is more ancestral, placentae incorporating the chorion and allantois (chorioallantoic placentation), as seen in eutherian mammals, are newer structures. These allowed relatively longer gestation periods, enabling development to occur with effective thermoregulation and protection from pathogens^{257,261}. The overlap of gene expression between the chorioallantoic placenta and marsupial mammary glands indicates a systemic departure from yolk-dependent nourishment with the emergence of placentation and lactation^{262,263}. The evolutionarily novel placenta, therefore, serves as a vital conduit for nutrition from the mother to the fetus. The extraembryonic tissues and the placenta thus became a site at which the conflicting interests of the mother (to balance her own nutritional requirements and that of all the embryos in her litter) and the father (to maximize nutritional input to his offspring) could play out²⁶⁴. Given most alleles traverse across generations, this led to the evolution of a novel epigenetic mechanism, genomic imprinting, proposed to represent these conflicting interests.

Cells occupying the outside of the morula, epithelialize and form junctional complexes between each other to yield the trophectoderm (Fig. 1a; E3.5). This is the first exclusively extraembryonic cell type, from which many of the ectodermal components of the non-maternal portion of the placenta are later derived (Fig. 1b). At this stage, the cells in the inside, referred to as the inner cell mass (ICM), are pluripotent and contribute to all cell types of the embryo as well as the mesodermal and endodermal components of extraembryonic tissues.

Up until the morula-to-blastocyst transition, the transcription factors CDX2 and OCT4 (also known as POU5F1) (which later mark the trophectoderm and ICM, respectively) are ubiquitously expressed²⁶. Through mechanisms conserved between mammalian species, segregation of the trophectoderm and the ICM is facilitated by Hippo–YAP signalling²⁷. In outer cells, where Hippo signalling is off, the nuclear YAP–TEAD4 complex (TEAD4 is also known as TEF-3) activates expression of trophectoderm-specific transcription factors such as *Cdx2* and *Gata3* while repressing pluripotency-determining transcription factors such as *Sox2* and *Oct4* (refs. 26,28–31). However, in mammals such as ruminants, which require rapid trophectoderm proliferation for blastocyst formation, this repression can be delayed³². Within the ICM, where Hippo signalling is on, YAP is phosphorylated and consequently degraded, OCT4 continues to maintain a pluripotency-compatible state²⁷. These cell specification events are accompanied by morphogenetic processes that form the blastocyst – a mammalian novelty in which the blastocyst cavity replaces the yolk of their oviparous ancestors. Extraembryonic cells form a monolayer epithelium around the periphery, with the ICM positioned eccentrically within it. In the mouse, this occurs by E4.0 (ref. 33) (Fig. 1a). Mechanical forces have an important role in sculpting the blastocyst into its final form (reviewed elsewhere³⁴).

The second lineage segregation event happens within the ICM, at approximately, the 64-cell stage, and generates the epiblast and the PrE^{35–37}. This happens asynchronously, through FGF4 signalling-mediated modulation of the levels of homeobox protein NANOG and the transcription factor GATA6, until they become mutually exclusive and restricted to cells of epiblast and PrE fates, respectively^{35,36,38,39}. The ‘salt-and-pepper’ distribution of epiblast and PrE cells within the ICM is eventually resolved so that the PrE epithelializes and occupies a position adjacent to the blastocyst cavity. Sustained expression of FGFR2 helps to maintain the PrE fate^{37,40}. The Ephrin–Eph signalling pathway, which regulates differential cell adhesion and repulsion⁴¹ as well as mechanical differences in cell surface fluctuations and fluidity⁴², has been suggested as a sorting mechanism. Cells that do not segregate to their respective compartments are competitively eliminated through apoptosis^{43,44}.

At around the same time, trophectoderm cells surrounding the blastocyst cavity downregulate homeobox protein CDX2 and differentiate into mural trophectoderm (mTE)⁴⁵. A subset of peripheral PrE cells, near the mTE, undergo an epithelial–mesenchymal transition (EMT) to become the parietal endoderm (ParE), a fate maintained by GATA6 and SOX17 expression⁴⁶. ParE cells migrate around the blastocyst cavity, along the basement membrane of the mTE^{47,48} (Fig. 1a; E4.5). This is the first mesenchymal migratory event during mammalian embryogenesis and results in the ParE, together with the mTE, forming the parietal yolk sac. The mTE further differentiates to give rise to trophoblast giant cells, which aid in implantation (Fig. 1b). The trophectoderm overlying the epiblast, called polar trophectoderm (pTE), gives rise to trophoblast constituents of the ectoplacental cone and the extraembryonic ectoderm (ExE), from which placental components are later

derived^{49,50}. Early differentiation of the trophectoderm is necessary to facilitate implantation and to meet the nutritional needs of the embryo, whereas a pluripotent epiblast ensures that the capacity to respond to subsequent patterning cues is preserved.

Post-implantation fate of extraembryonic tissues

By the time of implantation, BMP signalling in the remaining PrE cells represses GATA6 and SOX17, thus preventing EMT in these cells, which instead express FOXA2 to give rise to the visceral endoderm⁴⁶ (Fig. 1a,b). This epithelial tissue, which envelops the epiblast and ExE^{51,52}, is an indispensable source of patterning cues, as we will describe subsequently. Following implantation at E4.5, ExE and epiblast cells proliferate and the mouse embryo starts to acquire its characteristic egg cylinder shape, with clear proximodistal asymmetry^{49,53}. A subset of epithelial visceral endoderm cells undergo a characteristic unidirectional migration from the distal end of the egg cylinder, thereby performing a critical role in patterning axial asymmetries in the underlying epiblast⁵⁴ (Fig. 1a; E5.5–E6.25 and see the ‘Organizers’ section of embryonic patterning)⁵⁴.

The next major event during which extraembryonic tissues are generated happens at E6.5 with the onset of gastrulation, the process through which the three definitive germ layers of ectoderm, mesoderm and endoderm are generated^{55,56}. The epithelialization of the epiblast commits its cells from a naive to a primed pluripotent state⁵⁷, capable of responding to the patterning cues from the visceral endoderm and ExE to undergo gastrulation⁵⁸. This process is marked by the formation of a transient structure known as the primitive streak, where epiblast cells undergo EMT and delaminate through the basement membrane, giving rise to the mesoderm and definitive endoderm⁵⁹ (Fig. 1). Cells of the epiblast that do not move through the streak differentiate along the ‘default’ pathway of contributing to the definitive ectoderm⁶. The first mesodermal cells to emerge from the streak migrate proximally, between the ExE and VE. Along with existing ectodermal and endodermal tissues, they contribute to various extraembryonic structures⁶⁰ from which the fetal membranes and placenta are derived (Fig. 1a,b; see the ‘Extraembryonic tissues in the gastrulating embryo’ section). Subsequent mesodermal cells migrating distally and laterally from the primitive streak give rise to various embryonic mesodermal lineages^{56,61} (Fig. 1b). By the end of gastrulation, the basic body plan of the embryo is established, endowing spatial information onto progenitor cell populations, so that they can correctly contribute to their respective organ primordia.

Epigenetics in development

Epigenetic information is encoded in the form of reversible modifications to DNA or to histones, controlling gene regulation and thus permitting the dynamic establishment of cell-type-specific patterns of gene activation or repression. Epigenetic modifications are heritable across cell divisions, providing a ‘memory’ of cell identity and state that is persistent through developmental time (Fig. 2). Epigenetic memory can also endure across longer time scales, for example, in the form of genomic imprinting, which is an inter-generational epigenetic signal transmitted from parent to offspring^{62,63} (Box 2). Consequently, ablation or mutation of epigenetic regulators in development can result in the loss of epigenetic inheritance and/or tissue-specific patterning, leading to developmental defects and often embryonic lethality (Fig. 2a). In the following sections, we discuss reprogramming of epigenetic marks and maintenance of genomic imprints in extraembryonic lineages compared with embryonic lineages and discuss how epigenetic marks can influence transcriptional regulation. In later sections, we

highlight extraembryonic structures that are disrupted by the loss of epigenetic marks in development.

Lineage-specific epigenetic dynamics in development

Epigenetic modifications are highly dynamic during early development, as the gametic signatures are almost completely erased to enable lineage-specific epigenetic patterning upon implantation¹⁰ (Fig. 2b). After fertilization, paternally inherited epigenetic marks from sperm are rapidly erased almost genome-wide through the active removal of DNA methylation and the replacement of protamines (proteins that densely package DNA within mature sperm) with maternal histones⁶⁴. Although active epigenetic marks inherited from the oocyte are similarly rapidly reprogrammed, repressive marks can persist on the maternal allele at non-imprinted loci until the blastocyst stage¹⁰. Active marks, such as histone 3 lysine 4 trimethylation (H3K4me3), are re-established on both maternal and paternal alleles during zygotic genome activation (when development comes under the control of products encoded by the embryonic genome rather than those inherited from the oocyte), where the deposition of H3K4me3 at active promoters is coupled to transcriptional activation^{65,66}. Conversely, both repressive marks (that is, DNA methylation, H3K9me3 and H3K27me3) and bivalent chromatin marks (that is, colocalized H3K4me3 and H3K27me3) are not re-established until after implantation^{67–70}. Notably, imprinted domains are protected from, or resistant to, these mechanisms and retain allelic repression through this reprogramming event (Fig. 2b and Box 2).

Although the post-implantation re-programming of repressive and bivalent chromatin marks and maintenance of imprinted loci appear to occur in parallel in both embryonic and extraembryonic lineages, there are striking lineage-specific patterns that emerge by gastrulation^{67,71,72} (Fig. 2b). In epiblast cells, DNA methylation is acquired almost genome-wide, except for a subset of regulatory regions that remain unmethylated⁷³. These regulatory regions, including promoters and enhancers, are protected from the *de novo* DNA methyltransferase machinery by the presence of H3K4me3 and H3K4me1, respectively^{74,75}. Conversely, in trophoblast-derived and PrE-derived cells, the genome acquires large blocks of partially methylated DNA, which includes a subset of CpG-rich promoters of developmental regulatory genes^{67,69,72}. This partially methylated state of trophoblast cell types is a conserved feature across mammals and is maintained throughout pregnancy⁷⁶. Its establishment is thought to be at least in part due to the antagonistic activity of *de novo* DNA methyltransferases and Polycomb repressive complexes⁷⁷.

In the transition from naive to primed pluripotency, epiblast cells also acquire a ‘super’-bivalent state at CpG-rich developmental promoters, with increased H3K4me3 and chromatin accessibility compared with bivalent promoters in visceral endoderm cells⁷⁸. This super-bivalent state occurs transiently during gastrulation and is proposed to enable rapid and coordinated activation of gene expression in response to subsequent developmental cues^{78–80}. The mechanisms underpinning the global epigenetic differences between embryonic and extraembryonic lineages remain to be explored. Furthermore, understanding how these patterns impact transcription factor binding and lineage-specific gene expression patterns will be important future questions.

Genomic imprinting of extraembryonic tissues

Genomic imprinting occurs at regions where a repressed epigenetic state is inherited from the gametes and maintained on one allele in a parent-of-origin manner, which then leads to mono-allelic expression

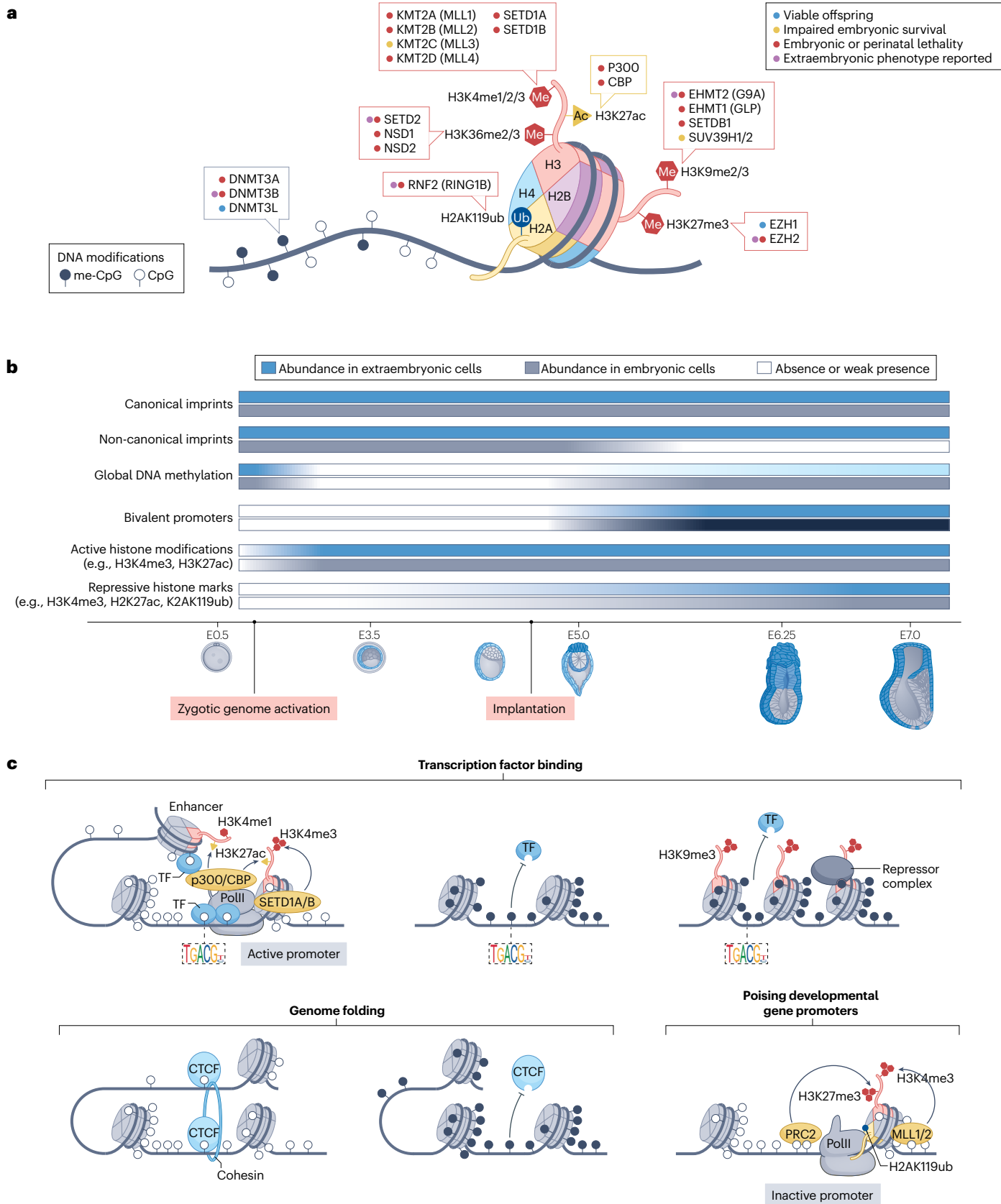


Fig. 2 | Role of epigenetics in development of embryonic and extraembryonic tissues.

a. Epigenetic modifications to DNA and its associated histones are programmed in development through the activity of epigenetic modifiers, which are denoted for the most well-studied modifications in the adjoining box. Knockout mouse models have shown that the majority of epigenetic modifiers are essential for development, and loss or mutation of these modifiers leads to embryonic or perinatal lethality or impaired embryonic survival. Although extraembryonic structures are not assessed in most models, several studies have reported extraembryonic phenotypes upon loss or mutation of epigenetic modifiers. **b.** Epigenetic modifications are dynamic throughout early development, as gametic patterns are erased in the pre-implantation embryo followed by re-establishment of embryonic and extraembryonic patterns. Canonical imprinted domains remain relatively static, with inherited allelic DNA methylation propagated through development in both embryonic and extraembryonic lineages. Conversely, non-canonical imprinted regions, including a subset of autosomal genes and the *Xist* gene in females, retain their imprinted status specifically in extraembryonic lineages, but, in embryonic lineages, are overwritten during post-implantation epigenetic reprogramming and hence lose their imprinting status. Globally, DNA methylation is largely erased during pre-implantation development and re-established post-implantation. Embryonic cells acquire high levels of DNA methylation, with the exception of regulatory regions, whereas extraembryonic cells acquire a partially methylated state across much of the genome. Bivalent promoters, which are enriched for both H3K4me3 and H3K27me3, are lost globally upon

fertilization and become re-established in the post-implantation embryonic and extraembryonic lineages. Importantly, embryonic cells transiently acquire a super-bivalent state with accentuated levels of H3K4me3 at these regions. Histone marks associated with active promoters, such as H3K4me3 and H3K27ac, are reprogrammed upon zygotic genome activation in a process dependent on the resumption of transcription. Repressive histone marks such as H3K9me3, H2AK119ub and H3K27me3 are erased in the pre-implantation embryo and are re-established around implantation. **c.** Epigenetic modifications can impact transcriptional regulation during development. Appropriate activation and looping of enhancers and promoters require the deposition of H3K4me1 and H3K4me3, respectively, by H3K4 methyltransferases and of H3K27ac by histone acetyltransferases. The presence of H3K4me at these regulatory regions excludes DNA methylation, facilitating binding of methylation-sensitive transcription factors (TFs). By contrast, DNA methylation of CpG-containing motifs (me-CpG) at silenced loci can sterically interfere with transcription factor binding. Accumulation of DNA methylation and H3K9me3, which are often colocalized, can recruit transcriptional repressor complexes and condense chromatin, causing the associated regulatory regions to be inaccessible to transcriptional activators. DNA methylation can also influence DNA looping through the inhibition of CTCF binding. The presence of a bivalent chromatin state at inactive developmental gene promoters can retain chromatin accessibility and paused polymerase II, allowing these regions to remain poised for activation by developmental transcription factors in response to the appropriate differentiation signals.

of associated genes⁸¹. Imprinting has evolved with the expansion of extraembryonic lineages in mammals⁸², and many imprinted genes have an important role in extraembryonic tissue development (discussed in Box 1). There are several examples of imprinted domains that are patterned distinctly between embryonic and extraembryonic cells⁸³, further illustrating that mechanisms of epigenetic programming are unique between the two lineages. Two forms of genomic imprinting have been described, canonical and non-canonical, both of which exhibit a lineage-specific effect. Canonical imprinted domains are repressed by allelic DNA methylation inherited from either sperm or oocyte, and these regions can often confer allelic expression of multiple surrounding genes, generating imprinted gene clusters⁸¹. Although canonical imprinted DNA methylation is largely maintained in both embryonic and extraembryonic lineages^{84,85} (Fig. 2b), the size of imprinted gene clusters can differ substantially. Expansion of a subset of imprinted gene clusters in extraembryonic lineages has been linked to allelic spreading of Polycomb and H3K9me2 (refs. 86–89); however, the mechanisms permitting this extended allelic repression are still not fully understood⁹⁰. Non-canonical imprinted domains are initially repressed by allelic H3K27me3 inherited from the oocyte, which then undergo an epigenetic switch to allelic DNA methylation in the transition from pre-implantation to post-implantation^{84,91,92}. However, non-canonical imprinting is only maintained in post-implantation extraembryonic lineages, whereas regions that were initially imprinted in this manner in the ICM lose this signature and become allelically indistinguishable in the epiblast^{91,93} (Fig. 2b). The mechanisms for the lineage-specific epigenetic patterning at non-canonical imprinted domains remain incompletely understood, but studies have demonstrated that it is essential for normal development^{94–96}.

Epigenetic influences on gene regulation

Once established, repressive epigenetic marks such as H3K9me3, H3K27me3 and DNA methylation can be inherited through cell divisions at imprinted or non-imprinted regions and serve as a memory

of the preceding chromatin state^{97,98}. These epigenetic states may be particularly relevant during the integration of differentiation signals (such as WNT, BMP and FGF) that are used in many developmental contexts, by limiting the accessibility of transcription factor binding sites, the responsiveness of regulatory elements and contacts between promoters and enhancers^{99,100} (Fig. 2c). Although each mark can exert its repressive function by reducing chromatin accessibility and recruiting repressor complexes^{101,102}, DNA methylation at transcription factor binding motifs can also directly sterically alter protein–DNA interactions. A screen of more than 500 human transcription factors revealed that for the majority, DNA methylation indeed either positively or negatively impacted binding affinity⁹⁹. Many of these transcription factors are essential for developmental processes. For example, methylation of the central CpG dinucleotide within the cyclic AMP response element (CRE) motif blocks binding of the ubiquitous transcription factor CREB1 and of cyclic AMP-dependent transcription factor ATF1, impacting target gene activation^{103–105}. Similarly, the transcription factor BANP showed a 16-fold reduced binding affinity to its CGCG-containing promoter-associated motif, if methylated¹⁰⁶. DNA methylation can also directly impact genome folding by inhibiting DNA binding of the insulator protein CTCF¹⁰⁷. CTCF can influence transcriptional regulation by altering DNA loop formation between enhancers and promoters, which has been clearly demonstrated for a number of genomic regions including the imprinted *H19–Igf2* locus^{107,108}. Importantly, disruption of *Igf2* expression in placenta results in impaired nutrient uptake and reduced fetal growth¹⁰⁹. Further studies investigating how epigenetics and transcription factors work together to instruct gene regulatory networks in different developmental contexts and, in particular, in extraembryonic cell types will be essential to understand (1) how lineage-specific epigenetic patterns impact the usage of transcription factor-binding sites, and (2) why the loss of epigenetic modifiers in vivo often results in remarkably cell-type-specific developmental phenotypes (discussed subsequently).

Implantation and shaping of the embryo

Implantation in eutherian mammals is the process by which the blastocyst contacts, adheres to and invades the endometrium of the uterus. It involves intricate molecular and cellular interactions between the blastocyst and the uterus, to ensure that the implantation competency of the former is closely synchronized with the receptivity of the latter. Implantation with a focus on maternal tissues is discussed in detail elsewhere^{110,111}. Here, we will outline the role of extraembryonic tissues

Box 2 | Evolution and function of genomic imprinting

Genomic imprinting is an epigenetic mechanism that facilitates the mono-allelic expression of several hundred genes based on parent of origin. Imprinting has evolved with placentation as evidenced by the fact that egg-laying monotremes show no evidence of imprinting and marsupials, with their relatively short gestational period, show imprinted expression of a smaller subset of genes compared with eutherian mammals²⁶⁵. The role of imprinted genes in placentation first became evident in early murine embryo manipulation experiments in which pronuclei were exchanged between newly fertilized zygotes to generate uniparental embryos. These studies demonstrated that parthenogenetic (maternal only) and androgenetic (paternal only) embryos failed to survive, showing either underdevelopment or overdevelopment of the trophoblast, respectively^{266,267}. Extensive identification and characterization of imprinted genes across mammals have since demonstrated their roles in regulating placental development, embryo growth, nutrient supply and maternal adaptation to pregnancy, with many, but not all, examples supporting the parental conflict hypothesis of imprinting^{109,268–270}. This postulates that during mammalian reproduction, the maternal and paternal genomes have opposing drives to regulate nutrient demand: maternal genes limiting fetal growth to ensure maternal survival and paternal genes promoting growth to ensure propagation of the patrilineage²⁷¹.

At the maternal–fetal interface, extraembryonic tissues can be influenced by both maternal and fetal signals and in turn regulate both nutrient acquisition from the mother and growth of the fetus. Hence, one would predict that extraembryonic tissues would be a primary site for genes to acquire novel imprinted status. Indeed, extraembryonic tissues exhibit many species-specific imprinted domains^{83,272,273}, supporting the idea that novel imprinted domains may continue to facilitate the adaptation of extraembryonic tissues to regulate nutrient allocation and fetal growth across species. Intriguingly, extraembryonic tissues in rodents uniquely exhibit a novel mechanism of imprinting (discussed in the ‘The role of epigenetics in the first extraembryonic tissues’ section), termed non-canonical imprinting, at a small subset of autosomal genes and at the *Xist* gene in females^{274,275}. Whether this form of imprinting exists across eutherian mammals or in marsupials remains unclear, but early evidence suggests that it is not conserved in human or pig^{168,276–279}. However, similar to extraembryonic tissues in rodents, marsupials have imprinted silencing of the paternal X chromosome in females²⁸⁰ and thus it remains an interesting future question whether this may be due to non-canonical imprinting of the marsupial *Xist*-like gene *Rsx*²⁸¹.

in implantation and in shaping the mouse embryo during the stages leading up to and directly after implantation.

Role of trophoblast derivatives in embryo implantation

Although the outermost layer forms a physical barrier between the embryo and its environment in other organisms, in eutherian mammals, the trophoblast of the blastocyst physically interacts with the maternal uterine lining to initiate implantation¹¹². The glycoprotein matrix surrounding the blastocysts, called zona pellucida, prevents premature ectopic implantation while the blastocyst moves through the oviduct¹¹³. Once in the uterus, the blastocyst ‘hatches’ from the zona pellucida, which is facilitated by expansion of the blastocyst cavity, thereby increasing the surface area of the embryo that can interact with the uterine endometrium¹¹⁴. The transcriptional profiles of mTE and pTE diverge rapidly following hatching^{115,116}. In the mouse mTE, the downregulation of YAP and TEAD4 in response to exposure to uterine oestrogen signals has been recently observed⁴⁵. Subsequent downregulation of CDX2 allows mTE cells to undergo EMT and differentiate into primary trophoblast giant cells – the first terminally differentiated cell type to arise during mouse embryogenesis¹¹⁷ (Fig. 1b). During this process, FGF signals produced by the epiblast protect the pTE from CDX2 downregulation, preventing their EMT and differentiation into trophoblast giant cells^{50,118–120}. mTE-derived trophoblast giant cells then make contact with the endometrial extracellular matrix through integrin–receptor interactions. The resulting activation of calcium signalling, mediated through fibronectin and phospholipase-C, further reinforces these adhesive interactions¹²¹. The sites of implantation are determined by uterine WNT signalling¹²². Trophoblast giant cells acquire an invasive behaviour, sending projections through endometrial intercellular spaces and inducing apoptosis in cells embedded within the uterine mucosa¹²³. In turn, uterine stromal cells proliferate and deposit large amounts of extracellular matrix proteins, a process known as decidualization, which engulfs the implanting embryo into the uterine tissue^{124,125} (see the figure in Box 3; mouse).

Implantation remains a challenging area to study owing to the inaccessibility of the embryo at these stages and the difficulty in recapitulating this process in vitro. Single-cell analyses of both the trophoblast and the uterine tissue at the maternal–embryonic interface have started to increase our understanding of the molecular mechanisms underlying implantation^{126,127}. In addition, advances are being made to model implantation in both mouse and human embryos^{112,128–130}, which will enable us to visualize the dynamic events during this process and further investigate molecular mechanisms through experimental intervention.

Role of extraembryonic tissues in shaping the blastocyst and egg cylinder

Extraembryonic tissues have a crucial role not only in controlling the size of the embryo by nutritional facilitation but also in shaping it. During pre-implantation stages, the outer cells, which later become the trophoblast, help determine the shape of the blastocyst. These cells pump osmolytes in between cells, driving influx of water, which creates intercellular fissures that coalesce to form a single blastocyst cavity^{131–133}. The integrity of the trophoblast epithelium, maintained by tight junctions, helps contain the blastocyst fluid, thus maintaining the overall shape of the blastocyst^{134,135}. Changes to luminal pressure and cortical tension occur with the expansion of the blastocyst cavity and are detected by trophoblast cells through vinculin-mediated mechanosensing, which in turn determines their fate and number before implantation¹³⁶.

Box 3 | Differences between post-implantation mouse and human embryos

Although mouse and human embryos show broad similarity in the lineal relationships of tissues and the molecular genetic control of cell type specification, they show marked differences. Some of the differences in the origin of extraembryonic tissues are highlighted here. In primates, including humans, an extraembryonic mesoderm and amniotic ectoderm are specified soon after implantation, in contrast to the mouse, where they are formed after the onset of gastrulation (see the 'Extraembryonic tissues in the gastrulating embryo' section). The human amniotic ectoderm is derived from the epiblast, and BMP4 from the amnion helps specify extraembryonic mesoderm²⁸², which is understood to be derived from the hypoblast, the primate equivalent of the mouse visceral endoderm^{259,283,284}.

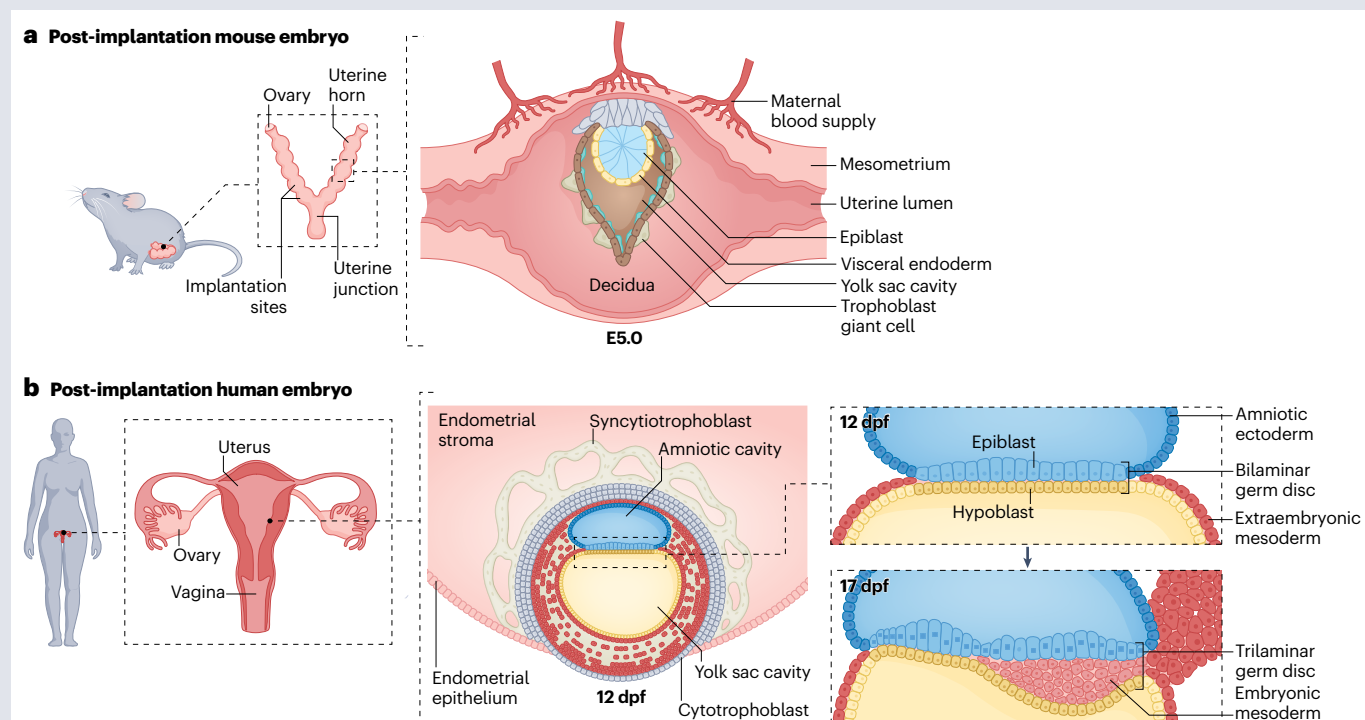
The top panel shows a post-implantation mouse embryo at embryonic day E5.0 in the context of maternal tissues. The embryo is enclosed within a decidua embedded within protective uterine tissue. In mice, implantation occurs around E4.5, and several conceptuses

can implant at the same time along the maternal uterine horns.

A single newly implanted embryo is shown, to indicate its orientation relative to the maternal tissues and blood supply.

The bottom panel shows a post-implantation human embryo from a comparable stage (Carnegie stage (CS) 5b, ~12 days post-fertilization (dpf)) in the context of the maternal tissues surrounding it.

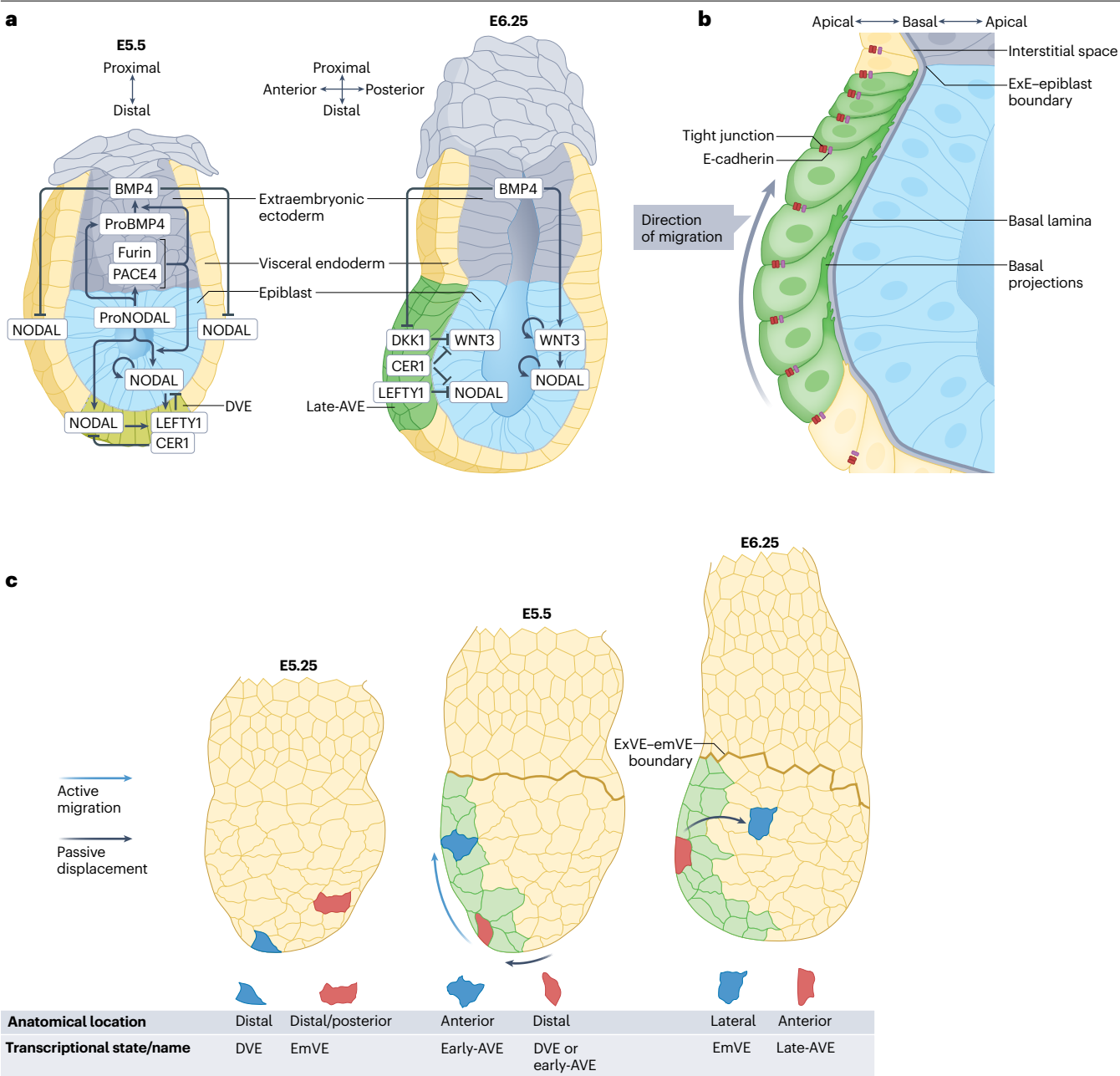
In contrast to the cup-shaped post-implantation mouse embryo, the human embryo at this stage is composed of a bilaminar disc, which is shown in higher magnification on the right, at 12 dpf (CS 5b). The bilaminar germ disc of the 12 dpf embryo is made up of the epiblast and the hypoblast. However, in contrast to the mouse, it also already possesses an amnion and extraembryonic mesoderm, before gastrulation. Embryonic mesoderm is generated only later, following gastrulation (~17 dpf; CS7), between the epiblast and the hypoblast, converting the bilaminar germ disc to a trilaminar germ disc. This would be roughly equivalent to the E7.0 embryo in Fig. 1a (the ~17-day human embryo illustration is adapted from ref. 285).



In the mouse embryo, the trophoblast also has a role in shaping the complex and polarized structure known as the egg cylinder (Fig. 1a; E4.5 onwards). This embryonic morphology is unique to specific rodents and is understood to enable these species to utilize the limited uterine space more effectively, allowing for much larger litter sizes (see the figure in Box 3; mouse). The adhesion of the trophoblast to the uterine matrix is necessary for the transition from the blastocyst to the egg cylinder¹¹². FGF signals from the epiblast help form the tissue boundary within the trophoblast allowing the pTE to expand and form the multilayered ExE¹³⁷. Ex vivo culture and imaging of implanting

stage embryos show that a release in tension in the pTE enables the formation of the elongated ExE¹³⁰ (Fig. 1a; E4.5). This coincides with the pluripotency state transition, a process driven by a switch in the genomic occupancy of OCT4 and global rearrangement of enhancer chromatin landscape to drive new transcriptional networks^{57,138}. Subsequently, tricellular tight junctions are formed between adjacent epiblast cells by the activation of the small GTPase RAP1, resulting in the de novo epithelialization of the epiblast¹³⁹.

The epiblast remains enveloped within the visceral endoderm, with the basal surfaces of both tissues in apposition, separated by a



common basement membrane (Fig. 1a). Reichert's membrane, a thick extracellular matrix deposited by the ParE cells of the parietal yolk sac onto the inner surface of the trophoctoderm, cushions the embryo against intrauterine contractions and has a role in further elongation of the egg cylinder in a controlled manner¹⁴⁰. Then the apical domains of epiblast cells cluster into a central rosette-like structure with high levels of CD34 family antiadhesins and low mechanical resistance^{133,141}. These apical contacts gradually open, resulting in the formation of the pro-amniotic cavity¹⁴¹ (Fig. 1a; E5.5).

The egg cylinder morphology of the mouse embryo could help shield more distal regions from the influence of proximal trophoctoderm-derived extraembryonic tissues by increasing the distance between them. Such a hypothesis has also been suggested

by comparative embryology. In embryos of many primates, including humans, similar shielding is achieved through the formation of an amniotic cavity before gastrulation¹⁴² (Box 3). Bovine embryos show a gradual loss of the trophoctoderm immediately abutting the epiblast. All these strategies lead to an attenuation in the influence of signals from trophoctoderm (or its derivatives) on the epiblast¹⁴³. In bovine embryos, signs of ectopic gastrulation are observed if the central epiblast maintains prolonged contact with the trophoctoderm, supporting this hypothesis¹⁴³. Therefore, despite the diversity in their shapes, mammalian embryos employ different strategies to ensure that the gene expression networks necessary to pass the critical checkpoints of development are activated at the right place and the right time.

Fig. 3 | The origin, function and fate of the distal and anterior visceral endoderm.

a, Key signalling pathways necessary for the induction of the distal visceral endoderm (DVE) at the distal tip of the E5.5 embryo are shown on the left. NODAL expression throughout the epiblast leads to expression of NODAL and LEFTY1 within the visceral endoderm and promotes expression of BMP4 in the extraembryonic ectoderm (ExE). BMP4 secreted by cells in the ExE suppresses NODAL expression in the proximal tissues of the egg cylinder. Therefore, NODAL expression in the visceral endoderm is most prominent in the region furthest away from the repressive BMP4 signal, that is, at the distal tip of the egg cylinder, enabling the formation of the DVE. LEFTY1 from the DVE acts as a secreted antagonist of NODAL signalling, which shifts epiblast NODAL expression to the more proximal aspects of the egg cylinder once the DVE is induced. Following DVE specification, its proximal migration leads to the establishment of an anterior visceral endoderm (AVE). The panel on the right shows the key signalling pathways required for the patterning function of the AVE from its anterior position at E6.5. As the AVE migrates, its expression of key NODAL and WNT inhibitors DKK1, LEFTY1 and CER1 is sustained. These suppress NODAL and WNT signalling in the epiblast immediately adjacent to the AVE, thereby restricting these 'posteriorizing' signals to the opposite side of the epiblast and establishing an anteroposterior axis orthogonal to the existing proximodistal axis. **b**, AVE cells send out basal projections in the direction of migration while maintaining epithelial integrity on their apical aspect through tight and adherens junctions (mediated by cadherin). Several modulators of actin dynamics are

phosphorylated differentially within AVE cells and could underpin the unique migratory behaviour of these in comparison to the surrounding embryonic visceral endoderm (emVE) cells. **c**, This panel illustrates the changing location and transcriptional state of visceral endoderm cells during DVE induction and AVE migration over a ~24-h period. The visceral endoderm changes dynamically across both space and time. DVE cells move from the distal tip of the egg cylinder to the exVE–emVE boundary. This process is driven by both the active migration and passive displacement of cells. Two examples of visceral endoderm cells are highlighted (red and blue), to illustrate how anatomical position and transcriptional states change during the course of AVE migration. The blue DVE cell specified at the distal tip migrates proximally towards the prospective anterior. Along its migratory path, the cell acquires the transcriptional state of the early-AVE. The red cell that later enters the distal domain follows a similar migratory trajectory, acquiring the transcriptional state of the more mature late-AVE, and by E6.25 occupies a position within the anterior region of the embryo. This incoming red cell and others displace the blue cell to the lateral regions of the embryo, where it downregulates its AVE-specific transcriptional programme to acquire one that resembles the surrounding emVE cells. This prevents AVE cells that are capable of suppressing the WNT and NODAL signals (required for primitive streak formation) from being close to the posterior of the embryo. The cells shaded green show the anatomical location of all cells comprising the early AVE and late AVE at E5.5 and E6.25, respectively.

'Organizers' of embryonic patterning

Biological systems as a whole are noisy, in comparison to the individual physical components they are built upon. Although groups of cells can autonomously self-organize into complex 3D structures *in vitro*^{11,144–146}, during development, subsets of cells act as organizers to determine the fate and morphology of other groups of cells^{147–149}. Organizers induce specific outcomes in nearby cells through signalling molecules that elicit complex cellular responses^{150,151}. These inductive signals can reinforce the intrinsic ability of neighbouring tissues to self-organize or add redundancy to the information provided by other processes, making the cellular mechanisms that coordinate patterning and morphogenesis more resilient to external noise or perturbations.

In mammalian embryos, the yolk sac continues to perform its ancestral role in the uptake, processing and transfer of nutrition to the developing embryo until a functional placenta is formed^{152,153}. Additionally, as we will explore, a subset of cells within the visceral endoderm (the tissue from which the yolk sac is later derived) acts as an organizer, providing an important source of patterning cues necessary for laying the foundations of the body plan. These are the cells of the distal visceral endoderm (DVE) and anterior visceral endoderm (AVE) that are responsible for establishing anteroposterior axial pattern in the embryo.

Induction of the distal visceral endoderm

As the egg cylinder elongates, the epithelial monolayer of the visceral endoderm is exposed to different signals along the proximodistal axis, depending on whether it is adjacent to the epiblast or ExE, leading to its specification into embryonic (emVE) and extraembryonic visceral endoderm (exVE), respectively (Fig. 1b). These two regions proceed to acquire different transcriptional signatures and become morphologically distinct; the emVE shows a squamous phenotype, whereas the exVE is cuboidal^{154–156}. Following the elongation of the egg cylinder at approximately E5.5, emVE cells at the distal tip undergo a characteristic thickening, becoming columnar, to form the DVE^{157–159}. The DVE is a

morphologically, transcriptionally and, later, behaviourally, distinct population of cells within the emVE that guides the establishment of anteroposterior asymmetry in the embryo^{156,158,160}.

DVE induction is initiated through the reciprocal interplay between the TGFβ superfamily members NODAL and BMP4 and their signal transducers SMAD2 and SMAD3, respectively^{161,162}. NODAL is expressed within the epiblast and induces *Nodal* and *Smad2* expression in the emVE itself^{163,164}. NODAL signals from the epiblast and visceral endoderm also regulate BMP4 signalling within another extraembryonic tissue, the ExE^{163,165} (Fig. 3a; E5.5). The ExE expresses the proprotein convertases Furin and PACE4 that help establish a gradient of active NODAL signal within the epiblast¹⁶⁶.

BMP4 from the ExE initially suppresses NODAL in the proximal regions of the emVE, thereby restricting DVE induction to the distal tip^{154,161,166,167} (Figs. 1a and 3a; E5.5). The arrangement of ExE and epiblast at the two ends of the elongating egg cylinder leads to the distancing of this distal region of the emVE from the source of the repressive signal¹⁵⁴. Furthermore, the localization of BMP receptors to the basolateral surface of epiblast cells helps grade the BMP response along the proximodistal axis, as the ligand, although present throughout the pro-amniotic cavity, can only enter the interstitial space to access the receptors through channels present at the ExE–epiblast boundary¹⁶⁸ (Fig. 3b).

There is likely to be interplay between chemical and mechanical cues in the induction of the DVE. At the distal tip of the egg cylinder, the visceral endoderm experiences maximum tissue curvature. Therefore, emVE cells at the distal tip are likely to be subject to increased mechanical stress, which could affect their fate and behaviour¹⁶⁹. The induction of the DVE, apparent from its columnarization, is accompanied by increased cortical tension as shown in a recent preprint (not yet peer-reviewed)¹⁶⁰. It remains to be determined whether the changes to the mechanical property of these cells is a consequence of, or a prerequisite to, their morphological transition. NODAL–SMAD2 signalling in the DVE eventually leads to the activation of the transcription factors LHX1 and FOXA2 that in turn activate or promote the existing

expression of NODAL pathway inhibitors *Cer1* and *Lefty1* (encoding CER1 and LEFTY1, respectively)^{159,164,170} (Fig. 3a; E5.5). These transcriptional changes in emVE cells mark the first emergence of proximodistal asymmetry within the embryonic half. The remarkable regionalization within a relatively small cohort of cells such as the emVE is maintained as cell mixing within the visceral endoderm is highly restricted, in contrast to the epiblast^{158,160,171,172}.

However, the cells of the visceral endoderm are not entirely static. They take part in a stereotypic cell migratory event that is essential to anterior patterning and the continued development of the embryo. Initial vital dye-labelling experiments showed that DVE cells move proximally along one side of the egg cylinder from their distal site of induction at E5.5, to occupy a position that will become the future anterior⁵⁴. Subsequent time-lapse live tracking using a DVE-specific fluorescent reporter showed that DVE cells are dynamic and migrate actively¹⁵⁸ (Fig. 3b). Once they initiate migration anteriorly from their original distal position, DVE cells are referred to as the AVE (Fig. 3a,b).

The relationship between the DVE and the AVE

The lineage relationship between the DVE and AVE can be a cause for misunderstanding. This is compounded by confusion arising from naming a population of migratory cells using anatomical terms such as ‘distal’ and ‘anterior’. The AVE exists in both the E5.5 (refs. 54,157,163,167,173,174) and E6.5 (refs. 175–180) egg cylinder, but is formed by different cell populations owing to ongoing migration. The AVE at both stages is characterized by similar marker gene expression, migratory behaviour and an asymmetric position within the emVE biased towards the prospective anterior^{156,158,174}. However, recent transcriptional profiling has shown that cells making up the ‘early-’ (E5.5) and ‘late-’ (E6.5) AVE, respectively, have distinct transcriptional signatures despite expressing similar markers¹⁵⁶ (Fig. 3c). It is therefore important to distinguish between these early-AVE and late-AVE populations and consider their respective relationships to the pre-migratory DVE induced at the distal tip.

Lineage-labelling studies show that the late-AVE (that is, emVE cells occupying the proximal anterior region of the embryo at E6.5) is not derived directly from the DVE (cells at the distal tip) of the E5.5 embryo¹⁵⁹ (Fig. 3c). This, however, should not be misunderstood to mean that the early-AVE, present at E5.5, is not derived from the DVE as it has been shown independently by both labelling^{157,178} and time-lapse microscopy^{158,174} that the DVE indeed gives rise to the E5.5 early-AVE. These studies also show that the DVE and early-AVE cells from the E5.5 embryo go on to occupy the proximolateral region in the E6.5 embryo, as a result of being displaced by new cohorts of AVE cells (the late-AVE) arriving between E5.5 and E6.5 at the anterior position from the distal egg cylinder (Fig. 3c).

Recent results from our group show the changing transcriptional state of these DVE and early-AVE cell populations as they migrate from the distal tip to anterior at E5.5 and from there to proximolateral regions¹⁵⁶. The early-AVE and the DVE from which it is derived are largely transcriptionally identical, differing only in expression levels of two genes, *Lefty1* and *Fgfbp1*. Similar to the DVE, the early-AVE continues to express factors essential for its axial patterning function (for example, *Cer1*; see the ‘The AVE and anteroposterior asymmetry’ section). As these early-AVE cells are displaced proximolaterally by E6.5, they sequentially downregulate their AVE transcriptional signature and revert to a ‘basal’ emVE transcriptional state¹⁵⁶ (Fig. 3c). This is essential to prevent these ‘former AVE’ cells from interfering with primitive streak formation, which occurs within the posterior regions of the

egg cylinder. By contrast, the late-AVE cells occupying the anterior of the E6.5 embryo now take on the patterning function of the ‘retired’ early-AVE, by continuing to express primitive streak repressors such as *Cer1* and *Dkk1*, as well as new markers that distinguish it from the early AVE such as *Cnn2*, *Frzb* and *Mical1* (ref. 156).

These findings show that the transition from DVE to AVE at E5.5 is a change in position of a population of migratory cells, rather than a differentiation event. Thus, the term ‘AVE’ should refer to the migratory population of emVE cells from E5.5 to E6.5, rather than only to a group of cells statically occupying the anterior position at E6.5. This takes into account that AVE migration is a dynamic process over a ~24-h period of time, during which cells from the distal tip move to occupy the anterior. It is not clear whether the late-AVE cells arrive at the anterior in a continuous manner behind the early-AVE, or whether the two populations migrate in distinct waves, with a pause in between – a question that more extensive time-lapse studies should resolve.

Directionality and mechanism of AVE migration

Most of our understanding of the process by which the AVE comes to acquire its anterior position derives from studies that focus on the migration of DVE cells to form the early-AVE. This migration occurs over a short window of developmental time within the geometric constraints of a curved epithelium, which imposes specific limits to its timing, speed, directionality, cohesivity, migratory end point and post-migration fate. It is necessary to closely coordinate AVE migration with changes of the molecular profile and mechanical landscape of the growing embryo. The cues that determine the direction of migration, what constitutes the end point of proximal AVE migration and the cellular mechanism of migration itself all remain open questions.

DVE cells take ~5 h to migrate from the distal tip of the egg cylinder up to the ExE¹⁵⁸. Once there, an as-yet unknown mechanism prevents these cells, now the early-AVE, from migrating beyond the epiblast onto the ExE. As a result, the first arriving early-AVE cells are displaced laterally by new, incoming late-AVE cells¹⁵⁸ (Fig. 3c). Cells of the exVE show regular packing and little-to-no neighbour exchange, presenting a non-permissive environment for the migration of AVE cells^{181,182}. However, the molecular basis for the differences in behaviour underpinning this barrier to AVE migration remains poorly understood. Marker expression specific to the exVE and emVE are indicators of an exVE–emVE gene expression boundary. Interestingly, these markers only align with the subjacent anatomical ExE–epiblast boundary in the prospective anterior. In the prospective posterior, the exVE marker expression domains extend more distally, into the VE overlying the epiblast¹⁵⁶. Differential mechanical properties and differential expression of cell–cell interaction mediators (such as ephrins, integrins, semaphorins or plexins) within these two subregions of the visceral endoderm could have a pivotal role in setting up this boundary to AVE migration^{115,156,160}. The characteristic pattern of AVE migration ensures that gene expression gradients initially aligned along the proximodistal axis gradually shift to be orthogonal to it, generating the prospective anteroposterior axis.

The requirement of the NODAL and WNT signalling pathways for AVE migration is demonstrated by migration arrest observed in mice carrying mutations for genes involved in these pathways, for example, *Cripto* and *Ctnnb* (encoding β -catenin), respectively^{183,184}. Mutants for *Lefty1* (a NODAL antagonist) also show migration defects, with the early-AVE abnormally migrating beyond the exVE–emVE boundary^{154,163,185,186}. The observed asymmetric expression of *Lefty1* and *Cer1* (a multivalent NODAL and WNT antagonist) at E5.5 has been

tracked back to their asymmetric expression in the pre-implantation PrE, when the blastocyst acquires bilateral symmetry with a distinct tilt in the positioning of its ICM derivatives^{159,187} (Fig. 1a; E4.0). However, direct evidence that pre-implantation expression of *Cer1* and *Lefty1* defines DVE precursor cells is lacking, and the asymmetry in transcript expression is not reflected at the protein level for CER1 and LEFTY1 (ref. 188). Other factors such as the WNT signalling inhibitor DKK1 can act as a guidance cue to migrating AVE cells when exogenously applied¹⁸⁹. However, although *Dkk1* in the embryo is expressed proximal to the migrating AVE, the timing of this localized expression is too late for it to act as the initial guidance cue⁵⁴. Furthermore, *Dkk1*-null embryos can gastrulate, calling into question its precise role during AVE migration¹⁹⁰.

In a recent preprint, we showed that patterned behavioural and biomechanical heterogeneities in the visceral endoderm facilitate, as well as restrict, DVE migration during the formation of the early-AVE¹⁶⁰. The exVE and cells of the posterior emVE remain static throughout AVE migration, whereas the rest of the emVE cells undergo global rearrangement to allow the AVE cells to migrate proximally^{160,181,191}. Proximal migration of early-AVE cells is an active process, characterized by basal projections in the direction of migration, whereas its displacement to the proximolateral aspects of the embryo is passive, the cells showing an absence of any projections¹⁵⁸ (Fig. 3b,c). Mutations in *Otx2* (which encodes a homeobox transcription factor expressed specifically in the AVE at these stages) show a global arrest of all emVE cellular movements, suggesting that active AVE migration results in cellular rearrangements in the surrounding emVE¹⁹¹.

Phosphorylation of cytoskeletal (for example, cyokeratin-8) and actin-binding proteins (such as MARCKS) is specific to the AVE and is thought to enable a finer control over migration by offering a mechanism for the quick turnover of the active proteome without having to rely on transcriptomic changes to alter cell behaviour¹⁵⁶. In addition, knockout studies have identified several modulators of actomyosin-based cytoskeletal dynamics and cell–cell interactions important for AVE migration: mutants of the *Rac1* gene (encoding the GTPase RAC1) fail to produce basal projections, whereas *βPix* (also known as *Arhgef7*, encoding Rho guanine nucleotide exchange factor 7) and *Sema6d* (encoding semaphorin-6d) mutants show defects in the directionality and length of the projections, respectively^{156,174,192}. Mutations leading to the loss of NAPI and PTEN (which are associated with the leading and trailing edge of migratory cells, respectively) cause early-AVE cells to disperse abnormally throughout the visceral endoderm^{193,194}. Such mutants in which AVE migration is defective without causing outright migration arrest enable us to better disentangle the molecular modulation of AVE migratory behaviour.

Epigenetic mechanisms have a role in regulating AVE migration. Embryos constitutively lacking the H3K4 mono- and di-methyltransferase MLL4 (also known as KMT2D) fail to appropriately specify the anteroposterior axis¹⁹⁵. *MLL4*-null embryos fail to correctly express key AVE-specific markers (including *Hhex*, *Dkk1*, *Lhx1* and *Lefty1*) and exhibit AVE migration defects¹⁹⁵. This is followed by a failure to localize *Bmp4*, *brachyury* (also known as *T*) and *Eomes* expression to the posterior ExE and epiblast¹⁹⁵. MLL4 is reported to be critical for enhancer activation during differentiation and cell fate transitions across multiple developmental contexts^{100,196–199}. The role for H3K4me1 and MLL4 has not been investigated in the differentiation of extraembryonic cell types; however, loss of MLL3 and MLL4 enzymatic function in embryonic stem cells resulted in impaired commitment to the extraembryonic endoderm lineage based on experiments using

induction (with activin A and retinoic acid) and embryoid body differentiation protocols²⁰⁰. This phenotype was linked to a disruption of GATA6 binding to enhancers. Thus, these data support the hypothesis that absence of MLL4 *in vivo* impairs appropriate rewiring of the gene regulatory network to facilitate the dynamic transcriptional response required for the AVE to transition from a static to a migratory population. Future work will be valuable to gain mechanistic insights into how loss of MLL4 and/or H3K4me1 specifically impacts transcription factor binding, gene activation and signalling within the embryo.

The AVE and anteroposterior axial asymmetry

In preparation for gastrulation, BMP from the ExE helps amplify *Nodal* and *brachyury* expression in the proximal epiblast, which is a prerequisite for primitive streak formation¹⁶⁶ (Fig. 3a; E6.5). However, it is the directional migration of the DVE that breaks the radial symmetry of gene expression within the epiblast⁵⁴. This is achieved by secretion of antagonists of WNT (DKK1 and CER1) and NODAL (LEFTY1 and CER1) pathways by migrating DVE cells^{161,163,176,201} and thereby downregulating posteriorizing signalling networks within the subjacent anterior epiblast (Fig. 3a; E6.5). This causes markers of the primitive streak (such as *Nodal*, *brachyury* and *Cripto*) to become asymmetric and restricted to the diametrically opposite side of the egg cylinder to where the AVE is established^{165,176,183,186} (Fig. 3a). For example, *Nodal* is initially expressed throughout the epiblast but upon induction of the DVE at the distal tip becomes restricted to the proximal epiblast^{162,179}. Subsequently, along with AVE migration to the prospective anterior, the *Nodal* expression domain shifts to the posterior epiblast, where it triggers the formation of the primitive streak^{156,163,180}. In the absence of NODAL, anterior-specific ectodermal genes are activated throughout the epiblast²⁰². The role of the AVE in correctly positioning the primitive streak is further demonstrated by mutants in which the AVE fails to be induced, does not function or shows impaired migration, resulting in the primitive streak being absent, disorganized, misplaced or duplicated²⁰³.

However, several questions remain about how the AVE restricts the formation of the primitive streak within the epiblast. There is considerable postmitotic cell mixing within the epiblast¹⁷², which is difficult to reconcile with a mechanism whereby spatially restricted repressive signals from the AVE have a localized influence on epiblast cells adjacent to the AVE. Moreover, embryonic stem-cell-derived aggregates cultured *in vitro* are able to asymmetrically express markers of the primitive streak even in the absence of an AVE^{204–206}, albeit with extremely low efficiency in comparison to embryos *in utero*. This suggests that there might be redundant mechanisms for the specification of axial asymmetry, given its fundamental importance in development. Considering the morphological constraints of the epiblast and how crowded its cells are within a pseudostratified epithelium, it is possible that any ability of these cells to self-organize is subject to considerable noise. On the basis of this, an intrinsic capability within the epiblast to generate axial asymmetry might be reinforced by the AVE to ensure reliable generation of axial asymmetry.

There is evidence from embryos developing *in utero* that cell movements within the epiblast are not entirely random: embryos chimeric for wild-type and *Nodal*-null cells show a significant enrichment of the *Nodal*-null cells specifically in the anterior¹⁷⁹. Directed cell movements within the epiblast would provide a cellular mechanism complementary to directional AVE migration for establishing anterior pattern. Such possibilities emphasize the need for a more detailed examination of the cell movements within the epiblast and how they may be coordinated with cell movements in the abutting visceral endoderm.

Extraembryonic tissues in the gastrulating embryo

Peri-gastrulation stages (E6.5–E8.0 in the mouse) are marked by extensive remodelling of extraembryonic tissues²⁰⁷ (Fig. 1a,b). Notably, at this stage, two extraembryonic tissues, the visceral endoderm and ExE, are thought to have important roles in specifying the number and localization, respectively, of primordial germ cells, the precursor cells that develop into germ cells, the first embryonic fate-restricted cell type²⁰⁸. In this section, we will outline how gastrulation generates the mesodermal components derived from the epiblast, which then coordinate with the existing extraembryonic ectodermal and extraembryonic endodermal lineages to form the fetal membranes as well as the placenta^{55,58,209} (Fig. 1b). As described earlier, by the time gastrulation is initiated, an anteroposterior axis is well established within the embryonic half of the conceptus in terms of gene expression and morphological asymmetries. However, within the extraembryonic half, proximodistal differences relative to the site of implantation remain the most pronounced axis of asymmetry, until mesodermal cells migrate into the extraembryonic half from the posterior side.

The AVE during gastrulation

Unlike in other vertebrates, it has been suggested that in mammalian embryos the head and trunk organizers are separate, with the latter function being exclusive to the node²¹⁰. The node is a transient structure that forms at the distal (or anterior) end of the primitive streak and serves as the gastrula organizer²¹¹. During gastrulation, visceral endoderm cells initially intercalate with the definitive endoderm^{7,52}, but by mid-gestation these cells appear to be eliminated from the gut endoderm through apoptosis²¹². This suggests a role for visceral endoderm cells immediately following gastrulation in patterning and morphogenesis. Indeed, explant experiments have shown that signals from the AVE are necessary for the induction of the heart, which is the most anterior structure of the mid-gastrulation embryo²¹³. The AVE is often referred to as the head organizer because genetic or mechanical removal of the AVE in gastrulating embryos results in the loss or reduction of forebrain markers^{175,178}. LHX1 and OTX2 form a transcriptional complex that is present at elevated levels in the anterior of the gastrulating embryo and activate expression of WNT antagonists²¹⁴. Chimeric embryos lacking the AVE-enriched transcription factors LHX1 or OTX2 in the extraembryonic tissues undergo gastrulation but show truncation of the forebrain, indicating that they are required for the neural patterning functions of the AVE^{179,215–217}.

Conditional ablation of *Bmp2* in the visceral endoderm only partially recapitulates the full *Bmp2*-null phenotypes, displaying only a subset of defects restricted to anterior structures – the head and the foregut²¹⁸. The post-gastrulation AVE has since been shown to be the source of BMP2 needed for the correct specification of these anterior structures²¹⁹. At later stages of gastrulation, many of these roles are taken on by epiblast-derived tissues – by the anterior mesendoderm and the anterior definitive endoderm – with crosstalk between these tissues helping to maintain anterior patterning within the neuroectoderm²²⁰. However, the AVE continues to have additional roles in coordinating morphogenesis within the anterior portion of the embryo, which are independent of its roles in anteroposterior patterning. Visceral endoderm-specific *Bmp2* mutants also show rostro-caudal defects, with ectopic neural folds and abnormal development of the heart in the exocoelomic cavity, highlighting a requirement of the AVE in directing embryo folding and positioning of the head structures and the heart^{218,219}.

Mesodermal contribution to extraembryonic tissues

Gastrulation marks the point at which the three definitive germ layers (ectoderm, mesoderm and endoderm) are specified and organized along the anatomical axes, to lay the foundations for various organ primordia. This involves concerted changes in cellular behaviours and fate specification. At the onset of gastrulation at approximately E6.5, a discrete population of epiblast cells on the proximal posterior side (diametrically opposite the AVE) delaminates to form the primitive streak, a region of disorganized epithelial integrity within the epiblast^{185,221–223}. This is orchestrated by several signalling pathways (WNT, BMP, NODAL and FGF) that act along precise spatiotemporal axial gradients set up partly by the AVE²²⁴ (Fig. 4a). Downstream of these signalling pathways, specific transcriptional programmes (mediated through *Mesp1*, *Twist*, *Crumbs2*, *Snail1/2* and *Zeb1/2*) that promote EMT are subsequently activated^{225–227} (Fig. 4b). Despite the different compositions of extraembryonic tissues between mammalian species and the signalling pathways involved, at the onset of gastrulation a core regulatory network of transcription factors is activated, leading to a conserved composition of cell states²²⁸. Recent evidence suggests that region-specific or graded differences in glucose metabolism along anatomical axes could coordinate with FGF signalling as a mechanism for fine-tuning the onset of gastrulation²²⁹. On the posterior side of the embryo, NODAL signalling regulates the expression of metalloproteinases that create perforations in the basement membrane, making it permissive to EMT^{223,230} (Fig. 4b). In the anterior, AVE cells secrete NODAL antagonists as discussed earlier and concurrently express proteins that help maintain basement membrane integrity outside the primitive streak²³¹.

Delaminating cells that emerge through the primitive streak migrate beneath the visceral endoderm towards the distal, anterior and proximal aspects of the egg cylinder and give rise to extraembryonic and embryonic mesodermal lineages and the definitive endoderm^{8,56,60} (Figs. 1a,b and 4c). Brachyury and FOXA2 are the master regulators of mesoderm and endoderm fates, respectively^{232,233}. However, although these fate decisions were originally thought to be binary, new evidence suggests that they are more complicated, with transcription factors working together combinatorially to allow lineage multifurcation and convergence events²³⁴. Over the next 12–24 h of development, the primitive streak extends distally and more cells are recruited along the streak^{207,223}. Lineage-labelling studies show that compared with the original position of the cells within the epiblast, their position along the primitive streak (posterior, middle and anterior) and the temporal order in which they are recruited (early, mid and late) are stronger determinants of their post-gastrulation fates^{58,60,211} (Fig. 4c). Exposure to localized morphogen landscapes dependent on the migratory paths taken by cells could also influence their behaviour^{173,235}. For example, recent data have suggested that coupling of an autoregulatory FGF loop with BMP repression could achieve differentiation of mesodermal subtypes in a spatially coherent manner²³⁶.

The first cells to emerge through the primitive streak at its most posterior end migrate proximally into the extraembryonic regions of the embryo^{237,238}, and show migratory behaviours that are distinct from the embryonic mesodermal cells emerging from the middle and anterior parts of the primitive streak. These nascent extraembryonic mesoderm cells migrate more slowly, with less directionality and through a GTPase-independent mechanism, compared with their embryonic counterparts²³⁸. The differences in behaviour could be due to differences in their cytoskeletal composition, the signalling molecules they are exposed to or the composition and mechanical

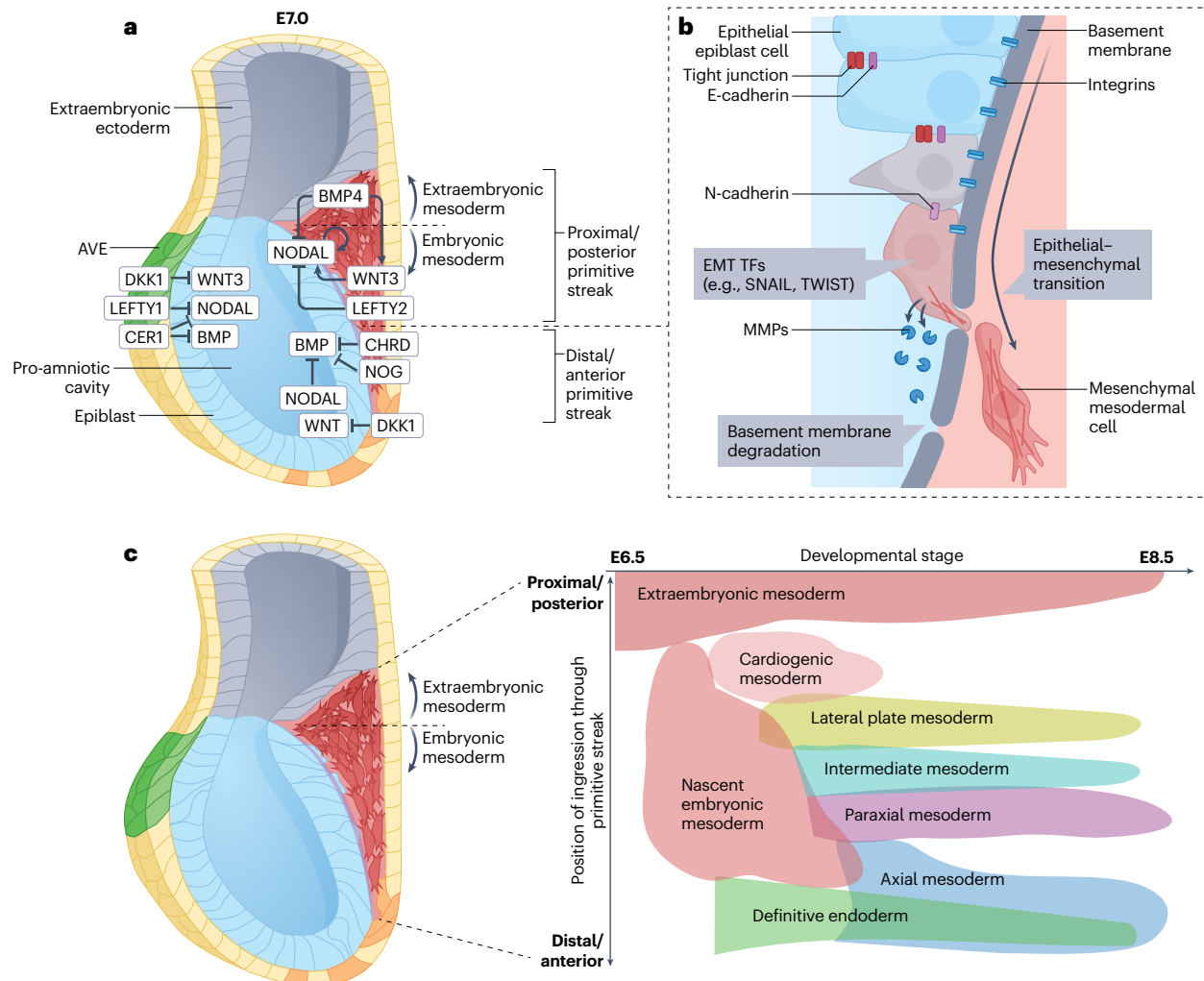


Fig. 4 | Signalling networks and cell lineages of the mouse gastrula. **a**, The critical signalling networks within the gastrulating embryo include NODAL, WNT, bone morphogenetic proteins (BMP) and fibroblast growth factors (FGF). Leading up to gastrulation, repressive signals from the anterior visceral endoderm (AVE) prevent signalling pathways that promote gastrulation from being induced in the anterior part of the embryo. Throughout the posterior proximal region of the epiblast, BMP, WNT, NODAL and FGF signalling remain active. This leads to the activation of transcription factors such as Brachyury, the key regulator of gastrulation, specifically in the posterior of the embryo. NODAL and BMP pathways inhibit each other and as a result, BMP4 is expressed proximally in the extraembryonic ectoderm (ExE) and the extraembryonic mesoderm, whereas NODAL is expressed more distally. The growth of the embryo moves distal cells away from the source of BMP. Cells within the distal primitive streak secrete the BMP antagonists chordin (CHRD), noggin (NOG) and the WNT antagonist DKK1, further restricting BMP and WNT signalling in distal regions. The proximal primitive streak is high in BMP and low in NODAL activity, an environment that promotes induction of extraembryonic mesoderm in the

adjacent tissue. By contrast, the distal primitive streak experiences low BMP and high NODAL, which promotes embryonic axial mesoderm and definitive endoderm fates. **b**, Molecular changes occur in proximal epiblast cells as they undergo epithelial–mesenchymal transition (EMT) during gastrulation. This involves the downregulation and disassembly of tight and E-cadherin-based adherens junctions, cadherin switching from E-cadherin to N-cadherin and secretion of metalloproteinases (MMPs) to modify the basement membrane. These phenotypical changes are driven by an EMT transcriptional programme mediated through transcription factors SNAIL and TWIST. **c**, Cells moving through the primitive streak commit to different fates based on the timing of ingression as well as the position along the primitive streak at which they delaminate. The first cells to emerge (through the posterior primitive streak) migrate proximally and give rise to extraembryonic lineages (extraembryonic mesoderm). Within the embryonic region, the distal-most primitive streak gives rise to the definitive endoderm and axial mesoderm. More proximally, the cells that emerge migrate towards the anterior midline giving rise to the cardiogenic mesoderm. Part **b** is adapted with permission from ref. 255, Elsevier.

properties of the extracellular environment the cells migrate through. How these different biochemical and biophysical parameters are integrated to determine differences in cell behaviour and fate are areas of ongoing study.

Extraembryonic mesodermal cells accumulate at the posterior ExE–epiblast boundary and form the amniochorionic fold, which grows to extend to the anterior midline. This forms the exocoelom, a cavity, which is delimited proximally by the chorion (formed by

extraembryonic mesoderm and ExE) and distally by the amnion²³⁷ (consisting of extraembryonic mesoderm and epiblast-derived amniotic ectoderm; Fig. 1b). Posterior epiblast cells act as the founding population for the amniotic ectoderm, but as the amniochorionic fold extends anteriorly, cells are also recruited from the proximal anterior epiblast^{239,240}. The walls of the exocoelom are lined by the visceral yolk sac (made from extraembryonic mesoderm and exVE), which later becomes the first site of haematopoiesis^{153,241,242} (Fig. 1b; blood islands). The primitive streak (marked by *brachyury* expression) extends proximally into the tissues surrounding the exocoelom and determines the site where the allantoic bud (precursor to the allantois) is induced²³². Although extraembryonic mesodermal cells transiting through the primitive streak form the allantois, fate mapping studies indicate that the streak-associated exVE can also contribute cells to the allantois²⁴³. Bipotency of the visceral endoderm, as discussed previously, is further evident at this point in mouse development, for example, in its contribution to the gut and yolk sac, both of which are of endodermal origin^{7,52}, but the possibility of cells crossing germ layers is an area that warrants further investigation.

Role of epigenetics in the formation of extraembryonic tissues in the gastrulating embryo

The morphogenesis and differentiation of extraembryonic mesoderm-derived structures, in addition to being regulated by signals from surrounding tissues, are also controlled by modulation of the epigenetic state of its cells. The constitutive loss of epigenetic repressors, including maintenance methyltransferase DNMT1, H3K9 methyltransferase EHMT2 (also known as G9A), HDAC1 and key components of the Polycomb repressive complex 1 (PRC1; for example, RNF2) and PRC2 (for example, EZH2, SUZ12 and EED), in mouse embryogenesis all lead to various extraembryonic malformations and embryonic lethality by E9.5 (refs. 244–253).

Loss of the PRC components EZH2, EED, SUZ12 or RNF2 leads to embryonic lethality shortly after gastrulation^{244–247}. Mesodermal cells in these embryos are abnormally misallocated to extraembryonic structures through excessive ingression at the proximal end of the primitive streak, resulting in both a reduced population of embryonic mesoderm and impaired formation of the chorion and amnion. Clonal analysis in early *Eed*-knockout mouse embryos showed that mesodermal cells exhibit a migration defect, failing to move anteriorly and hence accumulating in the extraembryonic compartment²⁴⁵. These misallocated mesodermal cells show a disrupted cell fate, originally evidenced by marker expression and recently confirmed using single-cell RNA sequencing in E8.5 *Eed* mutant embryos, which identified an overabundance of cells with ‘allantois’ identity^{248,249}.

Embryos lacking HDAC1 show disrupted cell proliferation, but undergo gastrulation, marked by *brachyury* expression, with mesoderm specification comparable to wild-type embryos²⁵⁰. However, formation of the allantoic extraembryonic mesoderm is severely impaired²⁵⁰. Embryos lacking EHMT2, which deposits H3K9me2, correctly form an allantois, but fail to undergo chorioallantoic attachment²⁵¹. Furthermore, abnormal allantoic outgrowth and a lack of blood in the yolk sac were reported in *Dnmt1*-null embryos²⁵². Indeed, depletion of primitive erythroid cells, among others, in *Dnmt1*-null E8.5 embryos was recently molecularly confirmed using single-cell RNA-sequencing²⁵³. Interestingly, this study also generated single-cell transcriptomic data from chimeric embryos using embryonic stem cells lacking DNA demethylase enzymes of the TET family. The TET triple-knockout cells were also unable to contribute to primitive haematopoiesis, despite

readily contributing to embryonic and extraembryonic mesoderm, demonstrating an innate requirement for dynamic DNA methylation patterning in this lineage. Thus, it is becoming more widely appreciated that an appropriate epigenetic state is required for the proper specification, differentiation and signalling-guided movement of extraembryonic mesoderm. Future advances in single-cell omics methods and complementary functional studies using in vitro systems, for example, embryoid body models for extraembryonic mesoderm differentiation, presented in a preprint²⁵⁴, will be key to our understanding of how epigenetics reshapes gene regulatory networks throughout this critical developmental window.

Concluding remarks

Signals are the sources of information that is passed between systems to produce or influence a specific outcome. Throughout the history of developmental biology, such signals have predominantly been considered to be molecules that mediate cell–cell communication – produced by a cell, triggering a downstream response within itself or another cell, changing their fate, behaviour or function. The advent and widespread use of multiplexed single-cell ‘omics’ approaches has provided us with a more coherent understanding of the changing molecular landscape during embryogenesis, at higher resolution (both spatial and temporal) and across scales of cells and populations of cells. Although secreted molecules have pivotal roles in relaying information between cells across short spatial and temporal ranges, epigenetic changes within cells act as a means through which memory of past signalling (and other) influences can be retained and even relayed across generations.

In addition to their indispensable roles in implantation and patterning discussed in this Review, extraembryonic tissues also perform nutritional roles in the fetus and even serve as a source of cells that perdure within the organism postnatally. Therefore, it is of utmost importance that these extraembryonic tissues are not disregarded but retained in single-cell multiomic studies, to interrogate the interplay between signals and epigenetics. Further refinement of in vitro embryo culture alongside advances in live-imaging technologies will help us better understand the cellular behaviours involving these tissues. Understanding extraembryonic tissues in this context will enable us to attain a more coherent understanding of the fundamental molecular and cellular mechanisms orchestrating embryonic development.

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